Network Traffic Analysis Report

- **Overview**

During the analysis of the captured network traffic between Kali Linux and Metasploitable 2, multiple insecure protocols were detected transmitting data in plaintext.

- **Findings**

- FTP Credentials Exposure
  Protocol: FTP (Port 21)
  Observed Credentials : USER anonymous
  PASS: IEUser@
- Issue: The FTP protocol transmits credentials in plaintext, making them easily accessible to attackers through network sniffing.
- Impact: Attackers can capture these credentials and attempt unauthorized access.
- Recommendation:* Avoid using FTP in production environments. Use secure alternatives such as SFTP or FTPS to protect sensitive data.

- Telnet Session Detected
  Protocol: Telnet (Port 23)
- Observed Behavior: Multiple Telnet session attempts were detected in the captured traffic, but no complete login credentials were recovered.
- Issue: Telnet transmits all session data in plaintext, including potential usernames and passwords.
- Impact: Any future login credentials sent over Telnet could be easily intercepted.
- Recommendation: Disable Telnet and use SSH (Secure Shell) instead for secure remote access.

- HTTP Traffic Analysis
  Protocol: HTTP (Port 80)
- Observed Behavior: HTTP requests were captured; however, no sensitive data such as usernames or passwords were observed in the capture.
- Issue: HTTP transmits data in plaintext by default.
- Recommendation: Ensure all web communications use HTTPS to prevent interception of sensitive data.

Conclusion

The captured network traffic revealed critical vulnerabilities related to the use of insecure protocols (FTP, Telnet, HTTP). Immediate action should be taken to migrate to secure alternatives to protect sensitive information from interception.

prepared by : abdelrhman salah